Peer-to-Peer Carpooling & Expense Splitter

Software Requirements Specification (SRS) - Lab 1

Problem Statement: A commuter ridesharing portal that matches drivers and passengers with overlapping daily routes, enforces detour tolerance thresholds, and calculates automated fuel cost share splits.
Key Actors: Rider Commuter, Driver Host

Requirements Summary Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall calculate route compatibility between driver and passenger commute paths, permitting matches only within a 1.5 km detour tolerance.	High	Given a driver's route and a passenger's pickup/dropoff points, the system returns a match if and only if the total added detour distance for the driver is $\le$ 1.5 km.	Ensures drivers are not inconvenienced by excessive detours while finding compatible passenger routes.
FR-002	Functional	The system shall calculate automated fuel cost share splits for matched trips based on total distance traveled, vehicle fuel efficiency, and current fuel prices, dividing the cost proportionally among occupants.	High	Upon completion of a shared ride, the system generates an itemized cost split bill showing total trip cost and each participant's exact share within 5 seconds.	Eliminates manual fare calculations and awkward negotiations by providing transparent, automated cost sharing based on objective parameters.
FR-003	Functional	The system shall allow Driver Hosts and Rider Commuters to set recurring daily commute schedules (e.g., weekday morning/evening pickup times) and query matching commute schedules.	High	Users can define recurring weekly time slots and receive match notifications for overlapping routes within a $\pm$15-minute departure window.	Enables regular daily commuters to establish consistent peer-to-peer carpooling routines without re-entering trip details daily.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-004	Functional	The system shall require Driver Hosts to submit valid driving licenses and vehicle registration documents for verification before publishing ride offers.	High	Unverified driver accounts are prevented from posting ride offers until document verification status is approved by the system or administrator.	Establishes baseline safety, trust, and accountability among peer-to-peer carpool participants.
FR-005	Functional	The system shall allow a Rider Commuter to send a ride request to a Driver Host for a compatible route and notify both parties upon acceptance or rejection.	Medium	Driver Hosts receive real-time notifications of ride requests and can accept or decline within a 30-minute window, updating seat availability automatically.	Facilitates direct peer-to-peer agreement and prevents overbooking of available vehicle seats.
NFR-001	Non-Functional	User ratings, driving license verification documents, and travel histories must be stored securely with encrypted backups.	High	Sensitive verification documents and travel logs are encrypted at rest using AES-256 and in transit using TLS 1.3, with automated daily encrypted backups.	Protects personally identifiable information (PII) and legal verification documents from data breaches and unauthorized access in compliance with privacy standards.
NFR-002	Non-Functional	The system shall compute route matching and detour tolerance calculations for incoming ride search queries within 2 seconds under a concurrent load of up to 1,000 active queries.	High	95% of route search queries return sorted match results within 2,000 milliseconds during stress testing at peak simulated load.	Ensures a responsive user experience so commuters can quickly evaluate matching rides during peak rush-hour usage.

Detailed Requirements Breakdown

Functional Requirements

FR-001: Route Compatibility & Detour Tolerance Matching

- **ID:** FR-001

- **Type:** Functional
- **Description:** The system shall calculate route compatibility between driver and passenger commute paths, permitting matches only within a 1.5 km detour tolerance.
- **Priority:** High
- **Acceptance Criteria:** Given a driver's route and a passenger's pickup/dropoff points, the system returns a match if and only if the total added detour distance for the driver does not exceed 1.5 km.
- **Rationale:** Ensures drivers are not inconvenienced by excessive detours while finding compatible passenger routes.

FR-002: Automated Fuel Cost Share Calculation

- **ID:** FR-002
- **Type:** Functional
- **Description:** The system shall calculate automated fuel cost share splits for matched trips based on total distance traveled, vehicle fuel efficiency, and current fuel prices, dividing the cost proportionally among occupants.
- **Priority:** High
- **Acceptance Criteria:** Upon completion of a shared ride, the system generates an itemized cost split bill showing total trip cost and each participant's exact share within 5 seconds.
- **Rationale:** Eliminates manual fare calculations and awkward negotiations by providing transparent, automated cost sharing based on objective parameters.

FR-003: Recurring Commute Schedule Matching

- **ID:** FR-003
- **Type:** Functional
- **Description:** The system shall allow Driver Hosts and Rider Commuters to set recurring daily commute schedules (e.g., weekday morning/evening pickup times) and query matching commute schedules.
- **Priority:** High
- **Acceptance Criteria:** Users can define recurring weekly time slots and receive match notifications for overlapping routes within a $\pm$15-minute departure window.
- **Rationale:** Enables regular daily commuters to establish consistent peer-to-peer carpooling routines without re-entering trip details every day.

FR-004: Driver Host License & Vehicle Verification

- **ID:** FR-004
- **Type:** Functional
- **Description:** The system shall require Driver Hosts to submit valid driving licenses and vehicle registration documents for verification before publishing ride offers.
- **Priority:** High
- **Acceptance Criteria:** Unverified driver accounts are prevented from posting ride offers until document verification status is approved by the system or administrator.

- **Rationale:** Establishes baseline safety, trust, and accountability among peer-to-peer carpool participants.

FR-005: Peer-to-Peer Ride Request & Confirmation Flow

- **ID:** FR-005
- **Type:** Functional
- **Description:** The system shall allow a Rider Commuter to send a ride request to a Driver Host for a compatible route and notify both parties upon acceptance or rejection.
- **Priority:** Medium
- **Acceptance Criteria:** Driver Hosts receive real-time notifications of ride requests and can accept or decline within a 30-minute window, updating seat availability automatically.
- **Rationale:** Facilitates direct peer-to-peer agreement and prevents double booking of available vehicle seats.

Non-Functional Requirements

NFR-001: Data Encryption & Secure Backup Storage

- **ID:** NFR-001
- **Type:** Non-Functional
- **Description:** User ratings, driving license verification documents, and travel histories must be stored securely with encrypted backups.
- **Priority:** High
- **Acceptance Criteria:** Sensitive verification documents and travel logs are encrypted at rest using AES-256 and in transit using TLS 1.3, with automated daily encrypted backups.
- **Rationale:** Protects personally identifiable information (PII) and legal verification documents from data breaches and unauthorized access in compliance with privacy standards.

NFR-002: Real-time Route Matching Performance

- **ID:** NFR-002
- **Type:** Non-Functional
- **Description:** The system shall compute route matching and detour tolerance calculations for incoming ride search queries within 2 seconds under a concurrent load of up to 1,000 active queries.
- **Priority:** High
- **Acceptance Criteria:** 95% of route search queries return sorted match results within 2,000 milliseconds during stress testing at peak simulated load.
- **Rationale:** Ensures a responsive user experience so commuters can quickly evaluate matching rides during peak rush-hour usage.